3.1 Instruction set encoding

Thumb instructions are either 16-bit or 32-bit. Bits[15:11] of the halfword that the PC points to determine whether it is a 16-bit instruction, or whether the following halfword is the second part of a 32-bit instruction.

Table 3-1 shows how the instruction set space is divided between 16-bit and 32-bit instructions. An x in the encoding indicates any bit, except that any combination of bits already defined is excluded.

Table 3-1 Determination of instruction length

hw1[15:11]	Function
0b11100	Thumb 16-bit unconditional branch instruction, defined in all Thumb architectures.
0b111xx	Thumb 32-bit instructions, defined in Thumb-2, see *Instruction encoding for 32-bit Thumb instructions* on page 3-12.
0bxxxxx	Thumb 16-bit instructions.

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3.2 Instruction encoding for 16-bit Thumb instructions

Figure 3-1 shows the main divisions of the Thumb 16-bit instruction set space. An entry in square brackets, for example [1], indicates a note below the table.

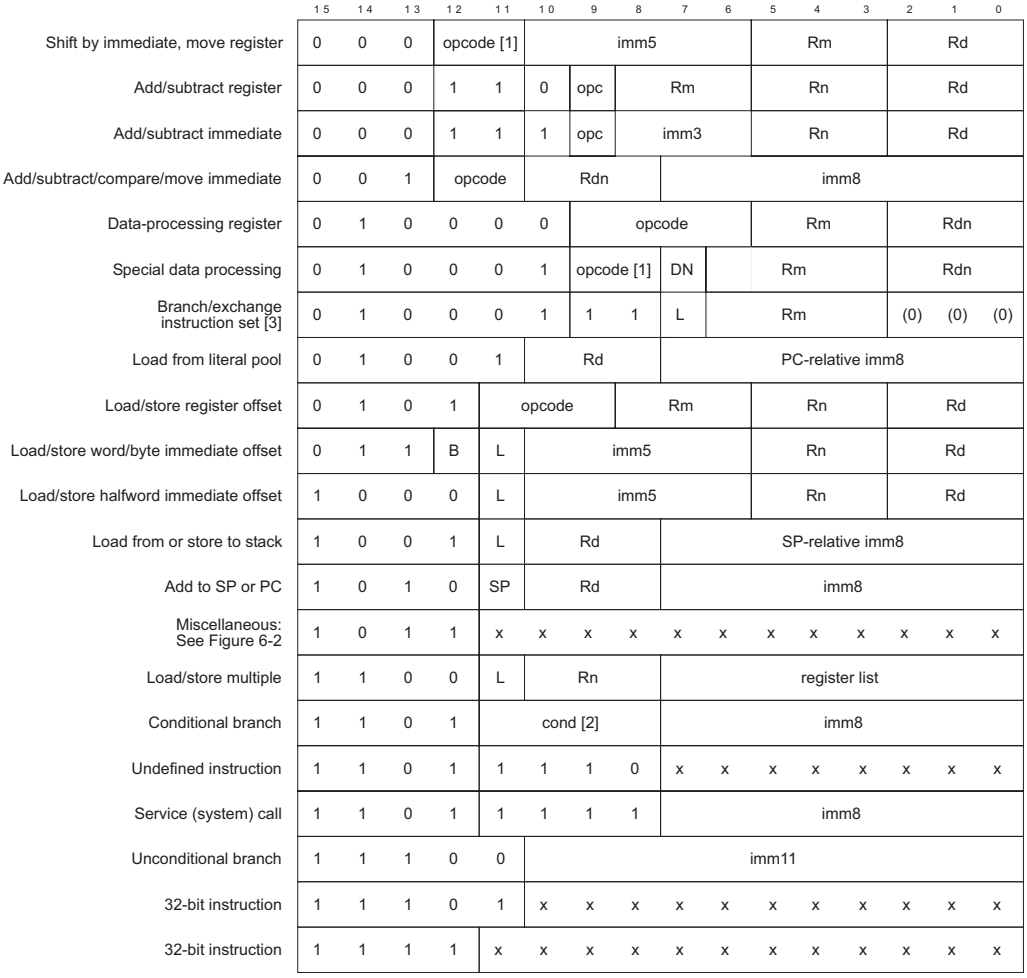


Figure 3-1 Thumb instruction set overview

1. opcode != 0b11.
2. cond != 0b111x.
3. The form with `L==1` is UNPREDICTABLE before ARMv5T.
4. This is an UNDEFINED instruction before ARMv5T.

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For further information about these instructions, see:

- Table 3-2 for shift (by immediate) and move (register) instructions
- Table 3-3 for add and subtract (register) instructions
- Table 3-4 on page 3-5 for add and subtract (3-bit immediate) instructions
- Table 3-5 on page 3-5 for add, subtract, compare and move (8-bit immediate) instructions
- Table 3-6 on page 3-6 for data processing (register) instructions
- Table 3-7 on page 3-6 for special data processing instructions
- Table 3-8 on page 3-7 for branch and exchange instruction set instructions
- *LDR (literal)* on page 4-102 for load from literal pool instructions
- Table 3-9 on page 3-7 for load and store (register offset) instructions
- Table 3-10 on page 3-7 for load and store, word or byte (immediate offset) instructions
- Table 3-11 on page 3-7 for load and store, halfword (immediate offset) instructions
- Table 3-12 on page 3-8 for load from or store to stack instructions
- Table 3-13 on page 3-8 for add 8-bit immediate to SP or PC instructions
- *Miscellaneous instructions* on page 3-9 for miscellaneous instructions
- Table 3-14 on page 3-8 for load and store multiple instructions
- *B* on page 4-38 for conditional branch instructions
- *SVC (formerly SWI)* on page 4-375 for service (system) call instructions
- *B* on page 4-38 for unconditional branch instructions.

Table 3-2 Shift by immediate and move (register) instructions

Function	Instruction	opcode	imm5
Move register (not in IT block)	*MOV (register)* on page 4-168	0b00	0b00000
Logical shift left	*LSL (immediate)* on page 4-150	0b00	!= 0b00000
Logical shift right	*LSR (immediate)* on page 4-154	0b01	any
Arithmetic shift right	*ASR (immediate)* on page 4-34	0b10	any

Table 3-3 Add and subtract (register) instructions

Function	Instruction	opc
Add register	*ADD (register)* on page 4-22	0b0
Subtract register	*SUB (register)* on page 4-367	0b1

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Table 3-4 Add and subtract (3-bit immediate) instructions

Function	Instruction	opc
Add immediate	*ADD (immediate)* on page 4-20	0b0
Subtract immediate	*SUB (immediate)* on page 4-365	0b1

Table 3-5 Add, subtract, compare, and move (8-bit immediate) instructions

Function	Instruction	opcode
Move immediate	*MOV (immediate)* on page 4-166	0b00
Compare immediate	*CMP (immediate)* on page 4-72	0b01
Add immediate	*ADD (immediate)* on page 4-20	0b10
Subtract immediate	*SUB (immediate)* on page 4-365	0b11

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Table 3-6 Data processing (register) instructions

Function	Instruction	opcode
Bitwise AND	*AND (register)* on page 4-32	0b0000
Bitwise Exclusive OR	*EOR (register)* on page 4-88	0b0001
Logical Shift Left	*LSL (register)* on page 4-152	0b0010
Logical Shift Right	*LSR (register)* on page 4-156	0b0011
Arithmetic shift right	*ASR (register)* on page 4-36	0b0100
Add with carry	*ADC (register)* on page 4-18	0b0101
Subtract with Carry	*SBC (register)* on page 4-261	0b0110
Rotate Right	*ROR (register)* on page 4-245	0b0111
Test	*TST (register)* on page 4-399	0b1000
Reverse subtract (from zero)	*RSB (immediate)* on page 4-249	0b1001
Compare	*CMP (register)* on page 4-74	0b1010
Compare Negative	*CMN (register)* on page 4-70	0b1011
Logical OR	*ORR (register)* on page 4-197	0b1100
Multiply	*MUL* on page 4-181	0b1101
Bit Clear	*BIC (register)* on page 4-46	0b1110
Move Negative	*MVN (register)* on page 4-185	0b1111

Table 3-7 Special data processing instructions

Function	Instruction	opcode
Add (register, including high registers)	*ADD (register)* on page 4-22	0b00
Compare (register, including high registers)	*CMP (register)* on page 4-74	0b01
Move (register, including high registers)	*MOV (register)* on page 4-168	0b10

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Table 3-8 Branch and exchange instruction set instructions

Function	Instruction	L
Branch and Exchange	*BX* on page 4-54	0b0
Branch with Link and Exchange	*BLX (register)* on page 4-52	0b1

Table 3-9 Load and store (register offset) instructions

Function	Instruction	opcode
Store word	*STR (register)* on page 4-339	0b000
Store halfword	*STRH (register)* on page 4-359	0b001
Store byte	*STRB (register)* on page 4-343	0b010
Load signed byte	*LDRSB (register)* on page 4-136	0b011
Load word	*LDR (register)* on page 4-104	0b100
Load unsigned halfword	*LDRH (register)* on page 4-128	0b101
Load unsigned byte	*LDRB (register)* on page 4-110	0b110
Load signed halfword	*LDRSH (register)* on page 4-144	0b111

Table 3-10 Load and store, word or byte (5-bit immediate offset) instructions

Function	Instruction	B	L
Store word	*STR (immediate)* on page 4-337	0b0	0b0
Load word	*LDR (immediate)* on page 4-100	0b0	0b1
Store byte	*STRB (immediate)* on page 4-341	0b1	0b0
Load byte	*LDRB (immediate)* on page 4-106	0b1	0b1

Table 3-11 Load and store halfword (5-bit immediate offset) instructions

Function	Instruction	L
Store halfword	*STRH (immediate)* on page 4-357	0b0
Load halfword	*LDRH (immediate)* on page 4-124	0b1

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Table 3-12 Load from stack and store to stack instructions

Function	Instruction	L
Store to stack	*STR (immediate)* on page 4-337	0b0
Load from stack	*LDR (immediate)* on page 4-100	0b1

Table 3-13 Add 8-bit immediate to SP or PC instructions

Function	Instruction	SP
Add (PC plus immediate)	*ADR* on page 4-28	0b0
Add (SP plus immediate)	*ADD (SP plus immediate)* on page 4-24	0b1

Table 3-14 Load and store multiple instructions

Function	Instruction	L
Store multiple	*STMIA / STMEA* on page 4-335	0b0
Load multiple	*LDMIA / LDMFD* on page 4-98	0b1

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3.2.1 Miscellaneous instructions

Figure 3-2 lists miscellaneous Thumb instructions. An entry in square brackets, for example [1], indicates a note below the figure.

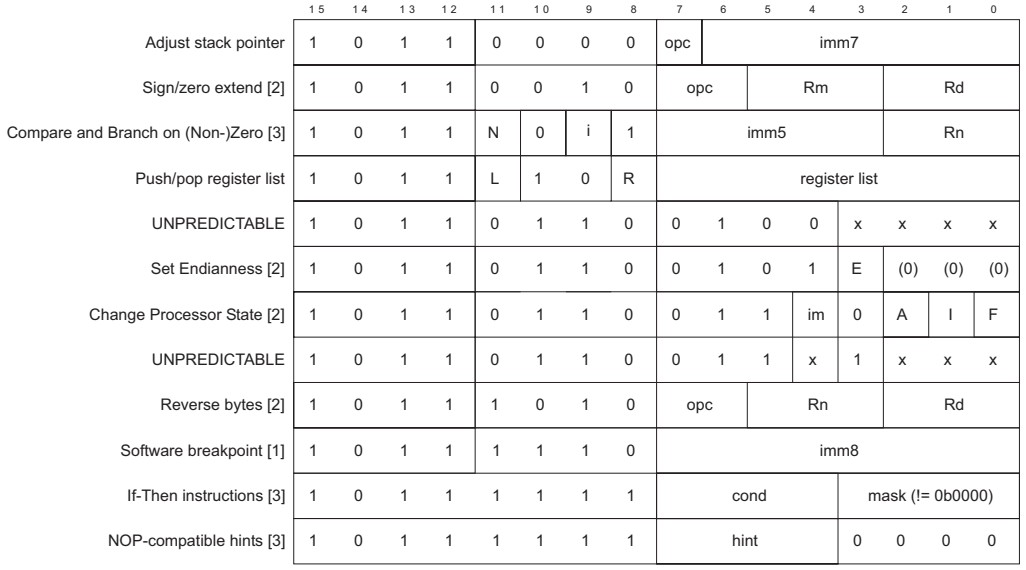


Figure 3-2 Miscellaneous Thumb instructions

1. This is an UNDEFINED instruction before ARMv5.
2. These are UNDEFINED instructions before ARMv6.
3. These are UNDEFINED instructions before Thumb-2.

Note

Any instruction with bits[15:12] = 1011, that is not shown in Figure 3-2, is an UNDEFINED instruction.

For further information about these instructions, see:

- Table 3-15 on page 3-10 for adjust stack pointer instructions
- Table 3-16 on page 3-10 for sign or zero extend instructions
- Table 3-17 on page 3-10 for compare (non-)zero and branch instructions
- Table 3-18 on page 3-10 for push and pop instructions
- *SETEND* on page 4-269 for the set endianness instruction
- *CPS* on page 4-76 for the change processor state instruction
- Table 3-19 on page 3-11 for reverse bytes instructions
- *BKPT* on page 4-48 for the software breakpoint instruction
- *IT* on page 4-92 for the If-Then instruction
- Table 3-20 on page 3-11 for NOP-compatible hint instructions.

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Table 3-15 Adjust stack pointer instructions

Function	Instruction	opc
Increment stack pointer	*ADD (SP plus immediate)* on page 4-24	0b0
Decrement stack pointer	*SUB (SP minus immediate)* on page 4-369	0b1

Table 3-16 Sign or zero extend instructions

Function	Instruction	opc
Signed Extend Halfword	*SXTH* on page 4-387	0b00
Signed Extend Byte	*SXTB* on page 4-383	0b01
Unsigned Extend Halfword	*UXTH* on page 4-465	0b10
Unsigned Extend Byte	*UXTB* on page 4-461	0b11

Table 3-17 Compare and branch on (non-)zero instructions

Function	Instruction	N
Compare and branch on zero	*CBZ* on page 4-60	0b0
Compare and branch on non-zero	*CBNZ* on page 4-58	0b1

Table 3-18 Push and pop instructions

Function	Instruction	L
Push registers	*PUSH* on page 4-211	0b0
Pop registers	*POP* on page 4-209	0b1

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Table 3-19 Reverse byte instructions

Function	Instruction	opc
Byte-Reverse Word	*REV* on page 4-235	0b00
Byte-Reverse Packed Halfword	*REV16* on page 4-237	0b01
UNDEFINED	-	0b10
Byte-Reverse Signed Halfword	*REVSH* on page 4-239	0b11

Table 3-20 NOP-compatible hint instructions

Function	Instruction	hint
No operation	*NOP* on page 4-189	0b0000
Yield	*YIELD* on page 4-471	0b0001
Wait For Event	*WFE* on page 4-467	0b0010
Wait For Interrupt	*WFI* on page 4-469	0b0011
Send event	*SEV* on page 4-271	0b0100

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